Programming and Databases

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Loops

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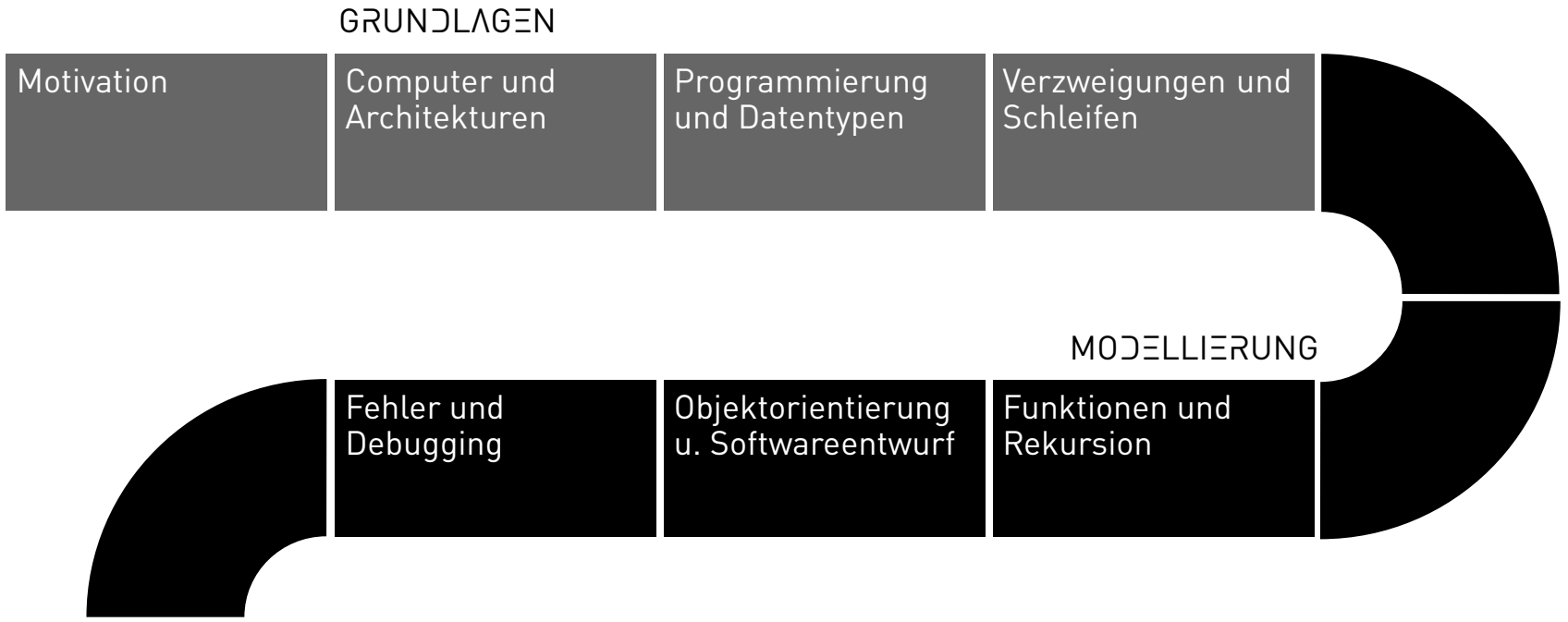
Process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Loops

Loops are an important approach in programming to implement repetitive or incremental solutions.
In Python, the following are available:

- `for` (equivalent to a for-each loop)
- `while` (equivalent to a while loop)

Important properties:

- They work exactly like all other control statements, using indentation for the block that follows the header line
- Unlike in functions, loop variables are also accessible outside the loop!
- The other loop variants are not explicitly present, but can be functionally emulated

For Loops - FOR Loop

The for loop repeats a block of statements for all elements in a sequence. Element is a variable that is always bound to the current element from the sequence.

```
for Element in Sequenz:
    # Statement
```

Loops - For-each Loop Example

The for loop in Python is a for-each loop. It always iterates through a sequence of values, such as a list. The iterator variable `e` contains the current value from the list `seq`.

```
seq = ['a','b','c']
for e in seq:
    print(e)
```

Output:

```
a
b
c
```

Loops - FOR-Loop Example

The for loop in Python can also be used as a traditional for loop, in which you repeat something n times. To do this, you generate a sequence of numbers with the `range(n)` function, through which the for-each loop then iterates.

```
n = 3
seq = range(n)
for e in seq:
    print(e)
```

Output:

```
0
1
2
```

Loops - FOR Loop Example

This can be used, for example, to convert a list of measurements from feet in the imperial system to metres.

```
measurements_feet = [4.2, 2.3, 6.2, 10.5]   # Input
measurements_meter = []   # Results

for feets in measurements_feet:
    measurements_meter.append(0.3048 * feets)

print(measurements_meter)
```

Output:

```
[1.2801600000000002, 0.70104, 1.88976, 3.2004]
```

Loops - WHILE Loop

The while loop runs as long as a condition is true. Since this is evaluated at the start and can be false from the outset, the body may not be executed.

```
while condition:
    # statement
```

As with an if statement, the condition is evaluated for its truth value:

- If this is `True`, the loop runs once, after which it is checked again
- If this is `False`, the loop ends and the code is not (re)executed

Loops - While Loop vs. Do-While Loop vs. Repeat-Until

While loop

In a while loop, the condition can be false right from the start. So the loop does not have to be executed.

```
Bedingung = True/False
while Bedingung:
    # Statement
    Bedingung = False
```

Do-While loop

In a do-while loop, the condition is checked only at the end. The loop is therefore always executed at least once.

```
Bedingung = True
while Bedingung:
    # Statement
    Bedingung = False
```

Repeat-Until loop

In a Repeat-Until loop, the condition is also checked only at the end. The loop is repeated until the condition becomes true.

```
Bedingung = False
while not Bedingung:
    # Statement
    Bedingung = True
```

Program Flow Control

We now know two tools:

Conditional execution:

- `if`, `else`, `elif`

Looping:

- `while`, `for`

Every program that has ever run on a computer could be written with these tools.

PROGRAMMIERUNG UND DATENBANKEN